KAIZEN MATHS

Lesson starter

Surds and Radicals

Fundamentals of Surds

Name _____

S1

Simplify: √32

S2

Simplify: √200

S3

Express √27 in the form a√b

S4

Simplify: √72

Show your working

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ALG-A602-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Surds and Radicals

Fundamentals of Surds

Name _____

E1

Simplify: √500

E2

Express √125 in the form a√b

E3

Simplify: √125

Show your working

KAIZEN MATHS

Exit ticket

Surds and Radicals

Fundamentals of Surds

Name _____

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Simplify: √500

E2

Express √125 in the form a√b

E3

Simplify: √125

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Show your working

KAIZEN MATHS

Exit ticket

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Simplify: √125

ALG-A602-L1 · E

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KAIZEN MATHS

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Show your working

Teacher answers & working

Surds and Radicals · Fundamentals of Surds

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S1

Simplify: $\sqrt{32}$

Answer:

$4\sqrt{2}$

Working:

Find a perfect square factor of 32

$32 = 16 \times 2$

Split the surd

$\sqrt{32} = \sqrt{16} \times \sqrt{2}$

Simplify the perfect square

$\sqrt{16} = 4$

$\sqrt{32} = 4\sqrt{2}$

S2

Simplify: $\sqrt{200}$

Answer:

$10\sqrt{2}$

Working:

Find the largest perfect square factor

$200 = 100 \times 2$

Split the surd

$\sqrt{200} = \sqrt{100} \times \sqrt{2}$

Simplify

$\sqrt{100} = 10$

$\sqrt{200} = 10\sqrt{2}$

Teacher answers & working

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S3

Express $\sqrt{27}$ in the form $a\sqrt{b}$

Answer:

$3\sqrt{3}$

Working:

Factor into a square and a remaining factor

$27 = 9 \times 3$

Write the square as a coefficient

$9 = 3^2$

Simplify

$\sqrt{27} = \sqrt{(3^2 \times 3)}$

$3\sqrt{3}$

S4

Simplify: $\sqrt{72}$

Answer:

$6\sqrt{2}$

Working:

Find a perfect square factor of 72

$72 = 36 \times 2$

Split the surd

$\sqrt{72} = \sqrt{36} \times \sqrt{2}$

Simplify the perfect square

$\sqrt{36} = 6$

$\sqrt{72} = 6\sqrt{2}$

Teacher answers & working

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E1

Simplify: $\sqrt{500}$

Answer:

$10\sqrt{5}$

Working:

Find the largest perfect square factor

$500 = 100 \times 5$

Split the surd

$\sqrt{500} = \sqrt{100} \times \sqrt{5}$

Simplify

$\sqrt{100} = 10$

$\sqrt{500} = 10\sqrt{5}$

E2

Express $\sqrt{125}$ in the form $a\sqrt{b}$

Answer:

$5\sqrt{5}$

Working:

Factor into a square and a remaining factor

$125 = 25 \times 5$

Write the square as a coefficient

$25 = 5^2$

Simplify

$\sqrt{125} = \sqrt{(5^2 \times 5)}$

$5\sqrt{5}$

Teacher answers & working

Surds and Radicals · Fundamentals of Surds

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E3

Simplify: $\sqrt{125}$

Answer:

$5\sqrt{5}$

Working:

Find a perfect square factor of 125

$125 = 25 \times 5$

Split the surd

$\sqrt{125} = \sqrt{25} \times \sqrt{5}$

Simplify the perfect square

$\sqrt{25} = 5$

$\sqrt{125} = 5\sqrt{5}$